FairnessOps: Metrics & Operational Definitions

Complete Guide to Fairness Audit Metrics

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1. Algorithm Bias (Demographic Fairness)

Plain English

Model makes systematically different predictions for different demographic groups with the same clinical features.

Operational Definition

For each protected attribute (race, gender, insurance):

1. Split patients into groups (Black vs White, Male vs Female, etc)
2. Get model predictions for each group
3. Compute metric for each group
4. Compare

Metrics

Option A: AUC Gap

```
AUC_White = 0.87
AUC_Black = 0.62
Gap = 0.87 - 0.62 = 0.25

Alert threshold: Gap > 0.10
```

Option B: Sensitivity Gap

```
Sensitivity_White = 0.85 (catches 85% of true sepsis cases)
Sensitivity_Black = 0.65 (catches 65% of true sepsis cases)
Gap = 0.20

Alert threshold: Gap > 0.10
```

Option C: Calibration Gap

```
Expected calibration error for White: 0.03
Expected calibration error for Black: 0.12
Gap = 0.09

Alert threshold: Gap > 0.05
```

What You Measure

```
For each demographic group:
  - AUC score
  - Sensitivity
  - Specificity
  - Calibration error

Then compute gap between groups
```

What You Do

```
Gap > threshold:
  ✗ ALERT: Model is biased
  ✓ Action: Investigate which features cause gap (SHAP)
  ✓ Action: Remove/adjust biased features
  ✓ Action: Retrain with fairness constraint
```

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2. Proxy Bias (Hidden Demographic Leakage)

Plain English

Model isn't directly using demographics, but it's using features that correlate with demographics.

Operational Definition

Train adversary to guess demographics from model predictions.

Metric

Adversary Accuracy

```
Can adversary predict race from predictions?
  Adversary accuracy: 0.72
  Baseline (random): 0.50
  Leakage strength: 0.72 - 0.50 = 0.22

Alert threshold: Leakage strength > 0.15
```

What You Measure

```
For each protected attribute:
  1. Train neural network to predict attribute from model output
  2. Measure accuracy on validation set
  3. Compare to baseline (random guessing)
  4. Compute leakage = accuracy - baseline
```

Interpretation

```
Leakage = 0.05: Minimal leakage (fair)
Leakage = 0.15: Moderate leakage (concerning)
Leakage = 0.30: High leakage (definitely biased)
```

What You Do

```
Leakage > 0.15:
  × ALERT: Model using demographic proxies
  ✓ Action: Use SHAP to identify which features leak
  ✓ Action: Remove proxy features
  ✓ Action: Retrain
```

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3. Feature Drift (Input Distribution Change)

Plain English

The characteristics of patients coming to the hospital are changing over time.

Operational Definition

Compare feature distributions between two time periods.

Metrics

Option A: Kolmogorov-Smirnov (KS) Test

```
Baseline period (Jan-Mar): WBC distribution
Current period (Apr-Jun): WBC distribution

KS statistic: 0.15 (how different are distributions?)
p-value: 0.002 (is difference statistically significant?)

Alert threshold: KS > 0.10 AND p-value < 0.05
```

Option B: Wasserstein Distance

```
Distance between baseline and current distribution: 0.25

Alert threshold: > 0.20
```

Option C: Population Stability Index (PSI)

```
For each feature:
  PSI = sum of (% in current - % in baseline) * ln(ratio)

  PSI < 0.10: Stable
  PSI 0.10-0.25: Minor shift
  PSI > 0.25: Major shift

Alert threshold: PSI > 0.10
```

What You Measure

```
For each clinical feature (WBC, age, lactate, etc):
  1. Get baseline distribution (e.g., Jan-Mar)
  2. Get current distribution (e.g., Apr-Jun)
  3. Compute KS or PSI
  4. Alert if > threshold
```

What You Do

```
Feature drift detected:
  × ALERT: Patient population is changing
```

```
  ✓ Action: Investigate why (season? policy change? new hospital wing?)
  ✓ Action: Retrain model on new distribution
  ✓ Action: Monitor if model performance degrades
```

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4. Algorithm Drift (Model Performance Change)

Plain English

Model's ability to make accurate predictions is getting worse over time.

Operational Definition

Track model's predictive performance over time.

Metrics

Option A: AUC Over Time

```
Jan: AUC = 0.85
Feb: AUC = 0.84
Mar: AUC = 0.82
Apr: AUC = 0.79
May: AUC = 0.76

Trend: Declining
Slope: -0.02 per month

Alert threshold: Slope < -0.01 (declining > 1% per month)
```

Option B: Sensitivity Over Time

```
Jan: Sensitivity = 0.88 (catches 88% of sepsis)
Feb: Sensitivity = 0.86
Mar: Sensitivity = 0.84
Apr: Sensitivity = 0.80

Trend: Declining
Alert threshold: Drop > 0.05 in any month
```

Option C: Calibration Over Time

```
Jan: ECE = 0.02 (well-calibrated)
Feb: ECE = 0.04
Mar: ECE = 0.07
Apr: ECE = 0.10

Trend: Getting worse
Alert threshold: ECE > 0.08
```

What You Measure

```
Weekly/Monthly:
  1. Compute AUC on new patients
  2. Compute sensitivity
  3. Compute calibration error
  4. Compare to baseline (at launch)
  5. Fit trend line (is it declining?)
```

What You Do

```
Algorithm drift detected:
  × ALERT: Model performance degrading
  ✓ Action: Investigate why (feature drift? concept drift? label quality?)
  ✓ Action: Retrain model
  ✓ Action: Check if still meets regulatory requirements
```

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5. Fairness Drift (Demographic Disparities Getting Worse)

Plain English

Model is becoming MORE biased over time.

Operational Definition

Track AUC gap / leakage over time.

Metrics

Option A: AUC Gap Over Time

```
Jan: Gap = 0.10 (fair)
Feb: Gap = 0.12
Mar: Gap = 0.15
Apr: Gap = 0.18
May: Gap = 0.22

Trend: Increasing (getting worse)
Slope: +0.03 per month

Alert threshold: Slope > +0.01
```

Option B: Proxy Leakage Over Time

```
Jan: Leakage = 0.10 (fair)
Feb: Leakage = 0.12
Mar: Leakage = 0.15
Apr: Leakage = 0.18

Trend: Increasing
Alert threshold: > 0.15 OR increasing > 0.02/month
```

What You Measure

```
Weekly/Monthly:
  1. Compute AUC gap (race, gender, insurance)
  2. Compute proxy leakage (for each attribute)
  3. Track over time
  4. Fit trend line
```

What You Do

```
Fairness drift detected:
  × ALERT: Disparity getting worse
  ✓ Action: Same as Algorithm Bias + investigate why
  ✓ Action: Did patient population change? (feature drift)
  ✓ Action: Did model change? (someone retrained it?)
```

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6. Label Bias (Outcome Measurement Inconsistency)

Plain English

Different demographic groups are being diagnosed/labeled inconsistently.

Operational Definition

Check if outcome is measured the same way across groups.

Metrics

Option A: Outcome Prevalence Gap

```
Prevalence of sepsis (confirmed cases) in:
  Black patients: 8%
  White patients: 12%

Gap: 4%

Question: Is this real (Black people get sepsis less)?
Or: Doctors not diagnosing sepsis in Black patients as often?

Alert threshold: Gap > 5% (investigate why)
```

Option B: Time-to-Diagnosis Gap

```
Average time from hospital admission to sepsis diagnosis:
  Black patients: 4.2 hours
  White patients: 2.8 hours

Gap: 1.4 hours

Alert: Black patients getting diagnosed later
       (maybe less monitoring? less clinical suspicion?)
```

What You Measure

```
For outcome (e.g., "sepsis diagnosed"):
  1. Compute prevalence in each demographic group
  2. Check time-to-event
  3. Compare across groups
  4. Investigate if gaps are medically justified
```

What You Do

```
Label bias detected:
  × ALERT: Outcome measured inconsistently
  ✓ Action: Review clinical protocols
  ✓ Action: Train clinicians on consistent diagnosis
  ✓ Action: Audit model is correctly trained
```

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7. Representation Bias (Small Sample Size for Groups)

Plain English

Some demographic groups don't have enough data in test set.

Operational Definition

Count patients in each demographic group.

Metrics

```
Test set (n=1000):
  White patients: 600 (60%)
  Black patients: 300 (30%)
  Asian patients: 100 (10%)

Minimum group size threshold: 100

Alert: All groups > 100 ✓ (okay)
       If Asian was 20: Alert × (too small to trust metrics)
```

What You Measure

```
For each demographic group:
  1. Count samples
  2. Compute percentage
  3. Check if > minimum threshold (e.g., 30 samples)
```

What You Do

```
Group too small:
  × ALERT: Can't reliably measure fairness for this group
```

```
  ✓ Action: Collect more data
  ✓ Action: Flag that metrics for this group are unreliable
  ✓ Action: Don't make claims about fairness for small groups
```

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8. Intersectionality Bias (Worst-Affected Groups)

Plain English

Some COMBINATIONS of demographics are more affected than others.

Operational Definition

Compute fairness metrics for combinations of attributes.

Metrics

```
Single attribute:
  Black patients AUC: 0.62

Intersectional:
  Black women AUC: 0.55 (worse)
  Black men AUC: 0.68 (better)
  Black women age 65+ AUC: 0.48 (much worse)

Alert: Identify worst-affected combination
       Focus resources there
```

What You Measure

```
For combinations (e.g., Black + Female + Age 65+):
  1. Get AUC for that specific group
  2. Rank all combinations by AUC
  3. Identify top 5 worst-affected groups
```

What You Do

```
Intersectionality identified:
  × ALERT: Black women 65+ has AUC 0.48 (vs 0.87 for White men)
  ✓ Action: Deploy separate model for this group
  ✓ Action: OR use group-specific thresholds
  ✓ Action: Prioritize this group for intervention
```

Summary Table: All Metrics

Bias Type	Metric	Measure	Alert Threshold
Demographic Fairness	AUC gap	White AUC - Black AUC	> 0.10
Proxy Bias	Leakage strength	Adversary acc - random	> 0.15
Feature Drift	KS statistic	Distribution change	> 0.10 + p<0.05
Algorithm Drift	AUC slope	Performance trend	< -0.01/month
Fairness Drift	Gap slope	Bias trend	> +0.01/month
Label Bias	Prevalence gap	Outcome difference	> 5%
Representation	Group size	N per group	< 30 samples
Intersectionality	Worst AUC	AUC for combinations	< 0.60

Implementation Priority

Start with (Essential):

1. **Demographic fairness** (easiest, most important)
2. **Proxy bias** (adversary) - this is your novel contribution
3. **Algorithm drift** (track over time)
4. **Fairness drift** (track disparities over time)

Then add (Important):

5. Feature drift
6. Label bias
7. Representation
8. Intersectionality

Example: Full Audit Output

```
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FAIRNESSOPS AUDIT REPORT
Model: sepsis_v1
Date: 2024-03-15
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1. DEMOGRAPHIC FAIRNESS
   AUC (White): 0.87
   AUC (Black): 0.62
   Gap: 0.25 → ALERT (> 0.10)

2. PROXY BIAS
   Leakage (Insurance): 0.45 → ALERT
   Leakage (Zip code): 0.38 → ALERT
   Leakage (Race): 0.18 → ALERT

3. ALGORITHM DRIFT
   Current AUC: 0.79
   Baseline AUC: 0.85
   Change: -0.06 → ALERT (degrading)

4. FAIRNESS DRIFT
   Current gap: 0.25
   Baseline gap: 0.10
   Trend: INCREASING → ALERT

5. LABEL BIAS
   Sepsis prevalence (White): 12%
   Sepsis prevalence (Black): 8%
   Gap: 4% → INVESTIGATE

6. REPRESENTATION
   Black patients in test set: 250 ✓ (sufficient)

7. INTERSECTIONALITY
   Worst group: Black women age 65+
   AUC: 0.48 → CRITICAL

RECOMMENDATIONS:
  1. Remove Insurance and Zip code features
  2. Retrain with fairness constraint
```

```
  3. Review sepsis diagnosis protocols
  4. Deploy separate model for Black women 65+
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```

Code Template: Computing Each Metric

Demographic Fairness

```python
def compute_auc_gap(y_true, predictions, demographics):
    """Compute AUC gap between demographic groups"""
    groups = demographics.unique()
    aucs = {}

    for group in groups:
        mask = demographics == group
        auc = roc_auc_score(y_true[mask], predictions[mask])
        aucs[group] = auc

    gap = max(aucs.values()) - min(aucs.values())
    return aucs, gap
```

Proxy Bias

```python
```

```python
def compute_leakage(predictions, demographics):
    """Train adversary to detect demographic leakage"""
    adversary = nn.Sequential(
        nn.Linear(1, 16),
        nn.ReLU(),
        nn.Linear(16, len(demographics.unique()))
    )

    # Train adversary...
    # Measure accuracy...

    accuracy = evaluate_adversary(adversary, predictions, demographics)
    baseline = 1.0 / len(demographics.unique())
    leakage = accuracy - baseline

    return leakage, accuracy
```

Algorithm Drift

```python
def detect_algorithm_drift(timestamps, aucs, window='month'):
    """Detect if model performance is degrading over time"""
    # Group by time window
    # Compute trend
    # Alert if slope < threshold

    slope = np.polyfit(range(len(aucs)), aucs, 1)[0]
    return slope, slope < -0.01
```

Notes

- All thresholds are suggestions; adjust based on clinical context
- Combine multiple metrics for robust assessment
- Validate findings with domain experts
- Document all decisions and alerts
- Update baselines when model is retrained

Generated: FairnessOps Metrics & Operational Definitions
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